Assignment - I Solutions

Astronomy & Astrophysics II

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Problem 1

The standard radiative transfer equation (RTE) is given by:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu$$

For the given 2D circular system, radiation of initial intensity $I_0 \equiv 1$ travels radially inward from the boundary $R_0 \equiv 1$ towards an observer at origin $r = 0$. For a fixed θ, the RTE is given by:

$$\frac{dI_\nu(r,\theta)}{dr} - \alpha_\nu(r,\theta)I_\nu(r,\theta) = j_\nu(r,\theta) \tag{1}$$

Since the RTE is a 1st order ODE, using the integrating factor $\mu(r,\theta) = \exp\left(-\int \alpha_\nu(r,\theta)dr\right)$, the general solution for the observed intensity $I_\nu(r,\theta)$ is given by:

$$I_\nu(r,\theta) = \frac{1}{\mu(r,\theta)}\left[-\int_1^r j_\nu(r',\theta)\mu(r',\theta)\,dr' + \mu(1,\theta)\right]$$

Case (a):

Given $j_\nu(r,\theta) \propto \frac{\cos^2\theta}{r}$, $\alpha_\nu(r,\theta) \propto r\cos^2\theta$. Assuming unit proportionality constants, the RTE is given by:

$$\frac{dI_\nu}{dr} - I_\nu r\cos^2\theta \;=\; \frac{\cos^2\theta}{r} \tag{2}$$

With $\mu(r) = e^{-\frac{1}{2}r^2\cos^2\theta}$, the integration results in an exponential integral $Ei(r,\theta)$. Hence, a numerical solution using `RK45` method is computed.

Case (b):

Given $j_\nu(r,\theta) \propto \cos\theta\sin\theta$, $\alpha_\nu(r) \propto re^{-r/R_0}$. Assuming unit proportionality constants, the RTE is given by:

$$\frac{dI_\nu}{dr} - I_\nu re^{-r} = \cos\theta\sin\theta \tag{3}$$

With $\mu(r) = \exp\left(e^{-r}(r+1)\right)$, the resulting integral does not have a closed-form solution in terms of elementary functions. Hence, a numerical solution using `RK45` method is computed.

Case (c):

Given $j_\nu(r,\theta) \propto r$, $\alpha_\nu(r,\theta) \propto r\ln(r/R_0)$. Assuming unit proportionality constants, the RTE is given by:

$$\frac{dI_\nu}{dr} - I_\nu r\ln r = r \tag{4}$$

With $\mu(r) = \exp\left(-\frac{r^2}{2}\ln r + \frac{r^2}{4}\right)$, the integration results in terms involving the error function $erf(x)$. Hence, a numerical solution using `RK45` method is computed.

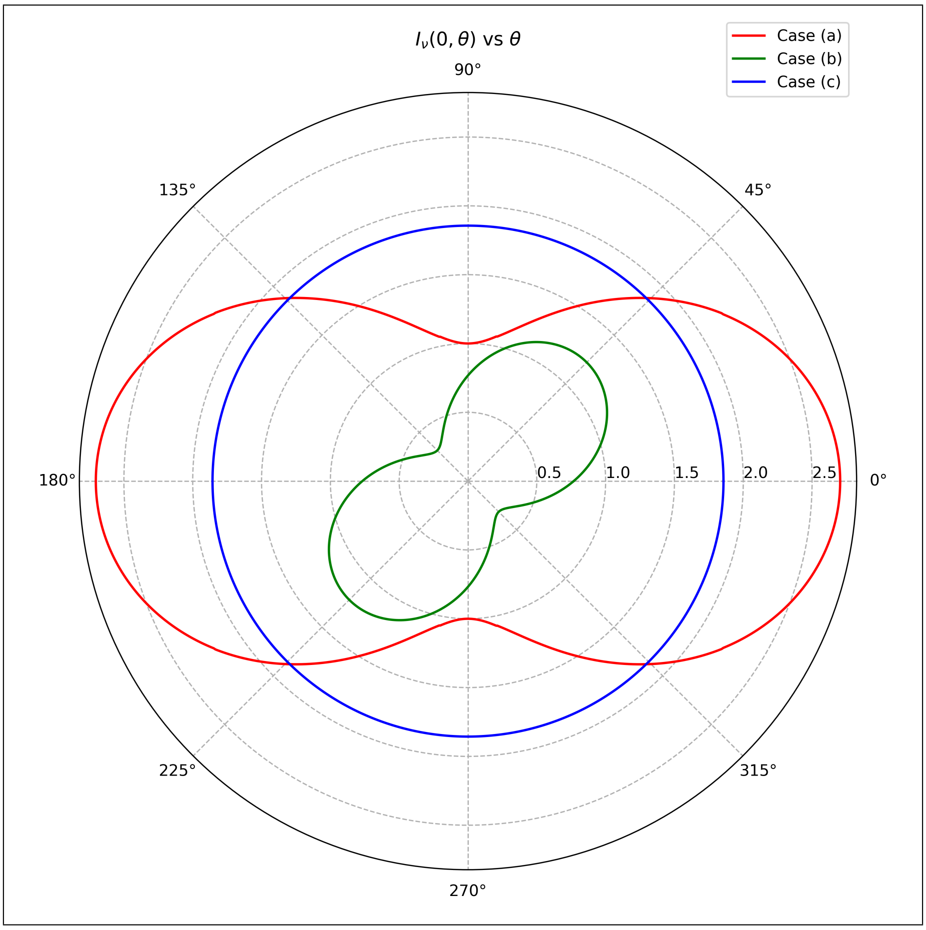

Figure 1: Normalised angular distribution of intensity at origin (Absorption & Emission).

`Link to code:` Question 1

Problem 2

Optical depth is given by:

$$\tau_\nu(s) = \int_s \alpha_\nu(s')ds'$$

For the given 2D circular system, the optical depth, measured radially inward from the boundary $R_0 \equiv 1$, is given by:

$$\tau_\nu(r,\theta) = -\int_1^r \alpha_\nu(r',\theta)dr' \tag{5}$$

The optical depth integrated over all angles is given by:

$$\tau_{int}(r) = -\int_0^{2\pi}\int_1^r \alpha_\nu(r',\theta)dr'd\theta \tag{6}$$

(i) Integrated Optical Depth

Assume unit proportionality constants.

- **Case (a):** $\alpha_\nu(r,\theta) = r\cos^2\theta$

$$\begin{aligned} \tau_{int}(r) &= -\int_0^{2\pi}\int_1^r r'\cos^2\theta\ dr'd\theta = \left(\frac{1-r^2}{2}\right)\int_0^{2\pi}\cos^2\theta\ d\theta \\ \implies \tau_{int}(r) &= \frac{\pi}{2}(1-r^2) \end{aligned} \tag{7}$$

- **Case (b):** $\alpha_\nu(r,\theta) \equiv \alpha_\nu(r) = re^{-r}$

$$\begin{aligned} \tau_{int}(r) &= -\int_0^{2\pi}\int_1^r r'e^{-r'}\ dr'd\theta = \left(\frac{r+1}{e^r} - \frac{2}{e}\right)\int_0^{2\pi} d\theta \\ \implies \tau_{int}(r) &= 2\pi\left(\frac{r+1}{e^r} - \frac{2}{e}\right) \end{aligned} \tag{8}$$

- **Case (c):** $\alpha_\nu(r,\theta) \equiv \alpha_\nu(r) = r\ln r$

$$\begin{aligned} \tau_{int}(r) &= -\int_0^{2\pi}\int_1^r r'\ln r'\ dr'd\theta = \left(\frac{r^2-1}{4} - \frac{r^2\ln r}{2}\right)\int_0^{2\pi} d\theta \\ \implies \tau_{int}(r) &= \pi\left(\frac{r^2-1}{2} - r^2\ln r\right) \end{aligned} \tag{9}$$

(ii) Angular Differential Optical Depth

Assume unit proportionality constants.

- **Case (a):** $\alpha_\nu(r,\theta) = r\cos^2\theta$

$$\begin{aligned} \tau(r,\theta) = \frac{1-r^2}{2}\cos^2\theta \quad &\implies \quad \frac{\partial\tau(r,\theta)}{\partial\theta} = -(1-r^2)\cos\theta\sin\theta \\ \left.\frac{\partial\tau(r,\theta)}{\partial\theta}\right|_{r=1/2} = -0.75\cos\theta\sin\theta \quad &\left.\frac{\partial\tau(r,\theta)}{\partial\theta}\right|_{r=1/4} = -0.9375\cos\theta\sin\theta \end{aligned} \tag{10}$$

- **Cases (b) & (c):** $\alpha_\nu(r,\theta) \equiv \alpha_\nu(r) = re^{-r}$, $\alpha_\nu(r,\theta) \equiv \alpha_\nu(r) = r\ln r$

  Since α_ν depends only on r, τ_ν is independent of θ.

$$\frac{\partial\tau(r,\theta)}{\partial\theta} = 0 \ \ \forall \ \ r \tag{11}$$

Problem 3

In the presence of both absorption and scattering in a medium, the RTE and optical depth are given by:

$$\frac{dI_\nu}{ds} = -\beta_\nu (I_\nu - S_\nu) \quad \text{where} \quad S_\nu = \frac{\alpha_\nu B_\nu + \sigma_\nu J_\nu}{\alpha_\nu + \sigma_\nu}$$
$$\tau_\nu(s) = \int_s \beta_\nu(s')ds'$$

Given $\sigma_\nu = 0.2\ \alpha_\nu$, the total extinction coefficient $\beta_\nu = \alpha_\nu + \sigma_\nu = 1.2\ \alpha_\nu$. Assuming $B_\nu \equiv J_\nu \equiv S_\nu$, the RTE and optical depth are given by:

$$\frac{dI_\nu}{ds} = 1.2\ \left(-\alpha_\nu I_\nu + j_\nu\right) \tag{12}$$
$$\tau_\nu(s) = 1.2 \int_s \alpha_\nu(s')ds' \tag{13}$$

(i) Observed intensity at origin

- **(A): $S_\nu \neq 0$**

$$\frac{dI_\nu(r,\theta)}{dr} = 1.2 \left\{-\alpha_\nu(r,\theta) I_\nu(r,\theta) + j_\nu(r,\theta)\right\} \tag{14}$$

Numerical solutions similar to Problem 1 are constructed using `RK45` method.

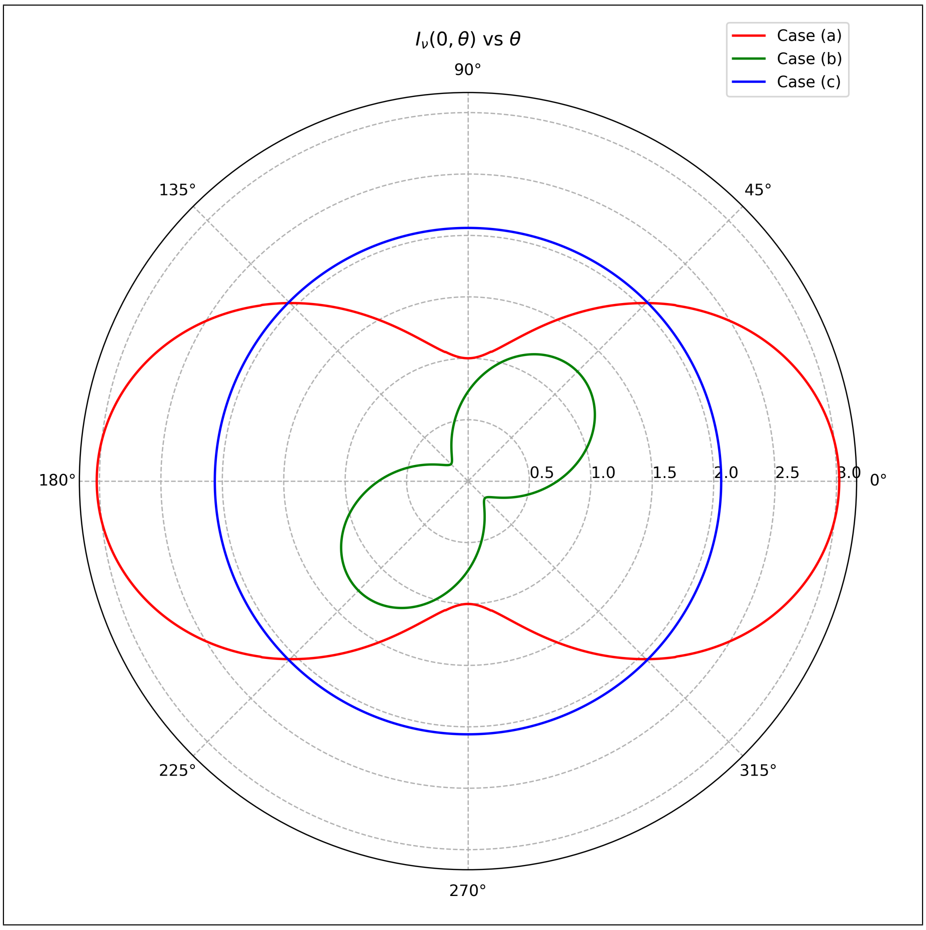

Figure 2: Normalised angular distribution of intensity at origin (Scattering & Emission).

- **(B): $S_\nu = 0 \implies j_\nu(r,\theta) = 0$**

$$\frac{dI_\nu(r,\theta)}{dr} = 1.2\ \alpha_\nu(r,\theta) I_\nu(r,\theta) \implies I_\nu(r,\theta) = e^{-\tau_\nu(r,\theta)} \tag{15}$$

  – **Case (a):**

$$I_\nu(r,\theta) = e^{-1.2\frac{1-r^2}{2}\cos^2\theta} \implies I_\nu(0,\theta) = e^{-0.6\cos^2\theta} \tag{16}$$

  – **Case (b):**

$$I_\nu(r,\theta) = e^{-1.2\left(\frac{r+1}{e^r} - \frac{2}{e}\right)} \implies I_\nu(0,\theta) = e^{-1.2\left(1-\frac{2}{e}\right)} \approx 0.728 \tag{17}$$

  – **Case (c):**

$$I_\nu(r,\theta) = e^{-1.2\left(\frac{r^2-1}{4} - \frac{r^2\ln r}{2}\right)} \implies I_\nu(0,\theta) = e^{0.3} \approx 1.350 \tag{18}$$

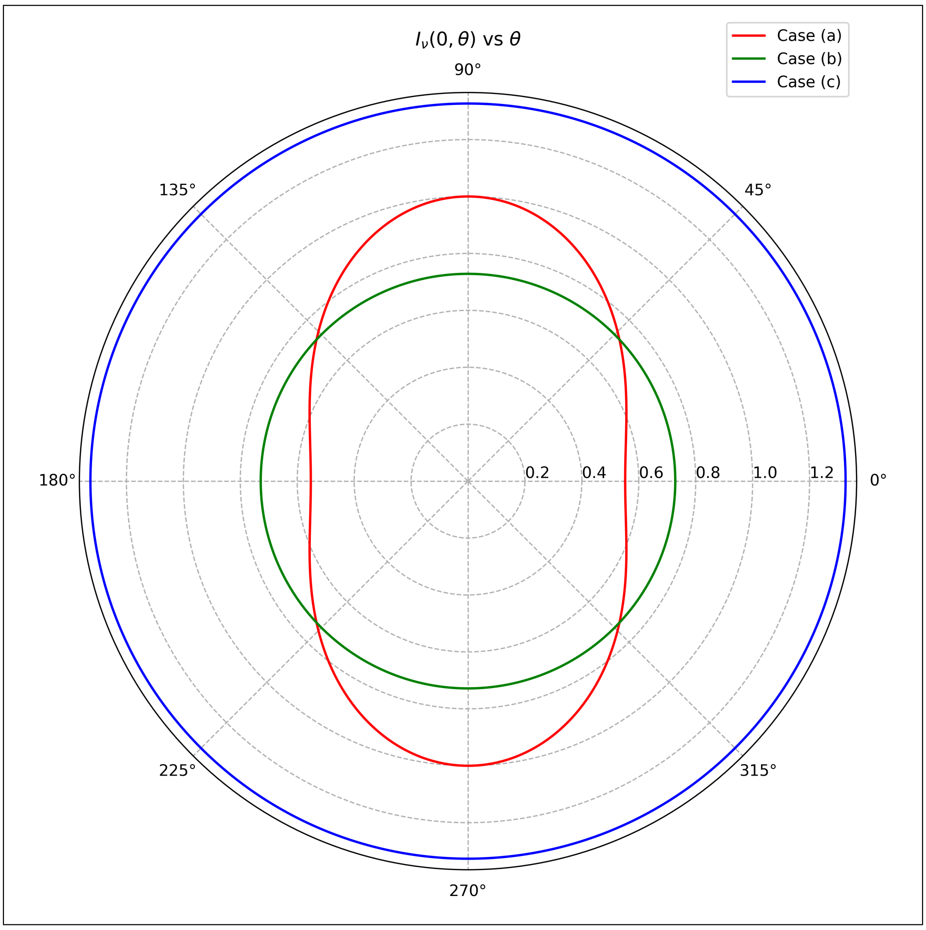

Figure 3: Normalised angular distribution of intensity at origin (Only Scattering).

(ii) Optical depth at origin

(A): $S_\nu \neq 0$ & (B): $S_\nu = 0 \implies j_\nu(r,\theta) = 0$

- **Case (a):**

$$\tau_\nu(r,\theta) = 1.2\frac{1-r^2}{2}\cos^2\theta \implies \tau_\nu(0,\theta) = 0.6\cos^2\theta \tag{19}$$

- **Case (b):**

$$\tau_\nu(r,\theta) = 1.2\left(\frac{r+1}{e^r} - \frac{2}{e}\right) \implies I_\nu(0,\theta) = 1.2\left(1 - \frac{2}{e}\right) \approx 0.317 \tag{20}$$

- **Case (c):**

$$\tau_\nu(r,\theta) = 1.2\left(\frac{r^2-1}{4} - \frac{r^2\ln r}{2}\right) \implies \tau_\nu(0,\theta) = 1.2\left(-\frac{1}{4}\right) = -0.3 \tag{21}$$

`Links to code:` Question 3(a), Question 3(b)

Problem 4

In the Rayleigh-Jeans regime, $I_\nu \propto T_b$, hence the brightness temperature obeys an equation equivalent to the RTE. For a medium at temperature $T \equiv 1000$ K with brightness temperature at boundary $T_{b,0}$, it is given by:

$$T_b(r,\theta) = T_{b,0}\ e^{-\tau_\nu(r,\theta)} + T\ (1 - e^{-\tau_\nu(r,\theta)}) = T + (T_{b,0} - T)\ e^{-\tau_\nu(r,\theta)} \tag{22}$$

(i) $T_{b,0} = 100$ K

$$T_b(0,\theta) = 1000 - 900e^{-\tau_\nu(0,\theta)} \tag{23}$$

- **Case (a):** $T_b(\theta) = 1000 - 900e^{-0.6\cos^2\theta} \implies \langle T_b \rangle \approx 318.177$ K
- **Case (b):** $T_b = 1000 - 900e^{-0.317} \approx 344.502$ K
- **Case (c):** $T_b = 1000 - 900e^{0.3} \approx -214.873$ K

`Note:` $T_b < 0$ in Case (c) is a direct consequence of $\alpha_\nu < 0$ for $r < 1$.

(ii) $T_{b,0} = 10000$ K

$$T_b(0,\theta) = 1000 + 9000e^{-\tau_\nu(0,\theta)} \tag{24}$$

- **Case (a):** $T_b(\theta) = 1000 + 9000e^{-0.6\cos^2\theta} \implies \langle T_b \rangle \approx 7818.226$ K
- **Case (b):** $T_b = 1000 + 9000e^{-0.317} \approx 7554.977$ K
- **Case (c):** $T_b = 1000 + 9000e^{0.3} \approx 13148.729$ K

Problem 5

The Liénard-Wiechert potentials of a single moving point charge Q are given by:

$$\Phi(\boldsymbol{r},t) = \left[\frac{Q}{R(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})}\right]_{ret}, \quad \boldsymbol{A}(\boldsymbol{r},t) = \left[\frac{Q\boldsymbol{\beta}}{R(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})}\right]_{ret}$$

where $R = |\boldsymbol{r} - \boldsymbol{r}'(t_{ret})|$ is the distance from the charge's retarded position to the observer, $\hat{\boldsymbol{n}}$ is the unit vector from the charge's retarded position to the observer, and $\boldsymbol{\beta} = \boldsymbol{v}/c$ is the charge's velocity. All the quantities are evaluated at a retarded time $t_{ret} = t - R/c$.

The resulting electromagnetic field is given by:

$$\boldsymbol{E}(\boldsymbol{r},t) = \left[\frac{Q}{R^2}\frac{(\hat{\boldsymbol{n}}-\boldsymbol{\beta})(1-\beta^2)}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})^3}\right]_{ret} + \left[\frac{Q}{Rc}\frac{\hat{\boldsymbol{n}}\times((\hat{\boldsymbol{n}}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}})}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})^3}\right]_{ret}$$

$$\boldsymbol{B}(\boldsymbol{r},t) = [\hat{\boldsymbol{n}}\times\boldsymbol{E}(\boldsymbol{r},t)]_{ret}$$

In the above electric field, the first term is the velocity field ($\sim 1/R^2$) and is a generalisation of the Coulomb field. The second term is the acceleration field ($\sim 1/R$) and contributes to the radiation, especially in the far-field approximation.

Assuming the charge undergoes non-relativistic motion, $\beta \to 0 \implies \hat{\boldsymbol{n}} - \boldsymbol{\beta} \approx \hat{\boldsymbol{n}},\ 1 - \boldsymbol{\beta}\cdot\hat{\boldsymbol{n}} \approx 1$, the radiation part of the field is given by:

$$\boldsymbol{E}_{rad}(\boldsymbol{r},t) = \frac{Q}{Rc^2}\,[\hat{\boldsymbol{n}}\times(\hat{\boldsymbol{n}}\times\boldsymbol{a})]_{ret}$$

$$\boldsymbol{B}_{rad}(\boldsymbol{r},t) = [\hat{\boldsymbol{n}}\times\boldsymbol{E}_{rad}]_{ret}$$

where $\dot{\boldsymbol{\beta}} = \boldsymbol{a}/c$ is the charge's acceleration. $\boldsymbol{E}_{rad}$, $\boldsymbol{B}_{rad}$, and $\hat{\boldsymbol{n}}$ form a mutually orthogonal right-hand triad, with $|\boldsymbol{E}_{rad}| = |\boldsymbol{B}_{rad}| = \frac{Qa}{Rc^2}\sin\theta$, where θ is the angle between $\boldsymbol{a}$ and $\hat{\boldsymbol{n}}$.

The Poynting vector and its magnitude are given by:

$$\boldsymbol{S} = \frac{c}{4\pi}(\boldsymbol{E}\times\boldsymbol{B}) = \frac{c}{4\pi}|\boldsymbol{E}_{rad}|^2\hat{\boldsymbol{n}} = \frac{c}{4\pi}|\boldsymbol{B}_{rad}|^2\hat{\boldsymbol{n}}$$

$$\implies |\boldsymbol{S}| \equiv \frac{dW}{dtd\Omega} = \frac{Q^2a^2}{4\pi R^2c^3}\sin^2\theta$$

The total power radiated is therefore given by:

$$P(\boldsymbol{r},t) \equiv \frac{dW}{dt} = \oint |\boldsymbol{S}|\ d\Omega$$

$$\implies P(\boldsymbol{r},t) = \frac{2Q^2a^2(\boldsymbol{r},t)}{3c^3} \tag{25}$$

For the given charge $Q = 4q$, Larmor's formula is given by:

$$P(\boldsymbol{r},t) = \frac{32q^2}{3c^3}a^2(\boldsymbol{r},t) \tag{26}$$

(i) Elliptic Trajectory

Given $x(t) = R_0\cos(\omega t),\ y(t) = R\sin(\omega t),\ z(t) = R$. a^2 is given by:

$$\ddot{x}(t) = -\omega^2R_0\cos(\omega t),\ \ddot{y}(t) = -\omega^2R\sin(\omega t),\ \ddot{z}(t) = 0$$

$$a^2(t) = \ddot{x}(t)^2 + \ddot{y}(t)^2 + \ddot{z}(t)^2$$

$$\implies a^2(t) = \omega^4\left\{R_0^2\cos^2(\omega t) + R^2\sin^2(\omega t)\right\}$$

Therefore, P is given by:

$$P(t) = \frac{32q^2\omega^4}{3c^3}\left\{R_0^2\cos^2(\omega t) + R^2\sin^2(\omega t)\right\} \tag{27}$$

(ii) Sine-Hyperbolic Trajectory

Given $x(t) = x_0 S(t)$ and $y(t) = y_0 S(t)$, where $S(t)$ is a sine-hyperbolic function. Defining the characteristic length of the system as $r_0 = \sqrt{x_0^2 + y_0^2}$, the trajectory is given by:

$$x(t) = x_0 \sinh\left(\frac{v_0 t}{r_0}\right), \; y(t) = y_0 \sinh\left(\frac{v_0 t}{r_0}\right)$$

where v_0 is the constant initial velocity. a^2 is given by:

$$\ddot{x}(t) = x_0 \left(\frac{v_0}{r_0}\right)^2 \sinh\left(\frac{v_0 t}{r_0}\right), \; \ddot{y}(t) = y_0 \left(\frac{v_0}{r_0}\right)^2 \sinh\left(\frac{v_0 t}{r_0}\right)$$

$$\begin{aligned} & a^2(t) = \ddot{x}(t)^2 + \ddot{y}(t)^2 \\ \implies & a^2(t) = \frac{v_0^4}{r_0^2} \sinh^2\left(\frac{v_0 t}{r_0}\right) = \frac{v_0^4}{x_0^2 + y_0^2} \sinh^2\left(\frac{v_0 t}{\sqrt{x_0^2 + y_0^2}}\right) \end{aligned}$$

Therefore, P is given by:

$$P(t) = \frac{32 q^2 v_0^4}{3 c^3 r_0^2} \sinh^2\left(\frac{v_0 t}{r_0}\right) = \frac{32 q^2 v_0^4}{3 c^3 (x_0^2 + y_0^2)} \sinh^2\left(\frac{v_0 t}{\sqrt{x_0^2 + y_0^2}}\right) \tag{28}$$

Problem 6

(i) Rosseland mean absorption coefficient

The Rosseland mean absorption coefficient is given by:

$$\frac{1}{\alpha_R} \equiv \frac{\int_0^\infty \frac{1}{\alpha_\nu+\sigma_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}$$

Assuming the absorption coefficient α_ν and the scattering coefficient σ_ν are independent of frequency, the Rosseland mean absorption coefficient is given by:

$$\alpha_R = \alpha + \sigma \tag{29}$$

- **Case (a): $\boldsymbol{\alpha_\nu = \alpha_0 r^2,\ \sigma_\nu = \sigma_0 r^2}$**

$$\alpha_R = (\alpha_0 + \sigma_0)\ r^2 \tag{30}$$

- **Case (b): $\boldsymbol{\alpha_\nu = \alpha_0 r,\ \sigma_\nu = \sigma_0 r}$**

$$\alpha_R = (\alpha_0 + \sigma_0)\ r \tag{31}$$

- **Case (c): $\boldsymbol{\alpha_\nu = \alpha_0 e^{-r/R_0},\ \sigma_\nu = \sigma_0 e^{-r/R_0}}$**

$$\alpha_R = (\alpha_0 + \sigma_0)\ e^{-r/R_0} \tag{32}$$

(ii) Rosseland energy flux approximation

The Rosseland energy flux approximation is given by:

$$F(r) = -\frac{16\sigma_S T^3}{3\alpha_R}\frac{\partial T}{\partial r}$$

where σ_S is the Stefan-Boltzmann constant. Since α_ν and σ_ν are assumed to be independent of frequency, the Rosseland energy flux approximation is given by:

$$F(r) = -\frac{16\sigma_S T^3}{3(\alpha + \sigma)}\frac{\partial T}{\partial r} \tag{33}$$

- **Case (a):** $\boldsymbol{\alpha_\nu = \alpha_0 r^2,\ \sigma_\nu = \sigma_0 r^2}$

$$F(r) = -\frac{16\sigma_S}{3(\alpha_0+\sigma_0)}\frac{T^3}{r^2}\frac{\partial T}{\partial r} \tag{34}$$

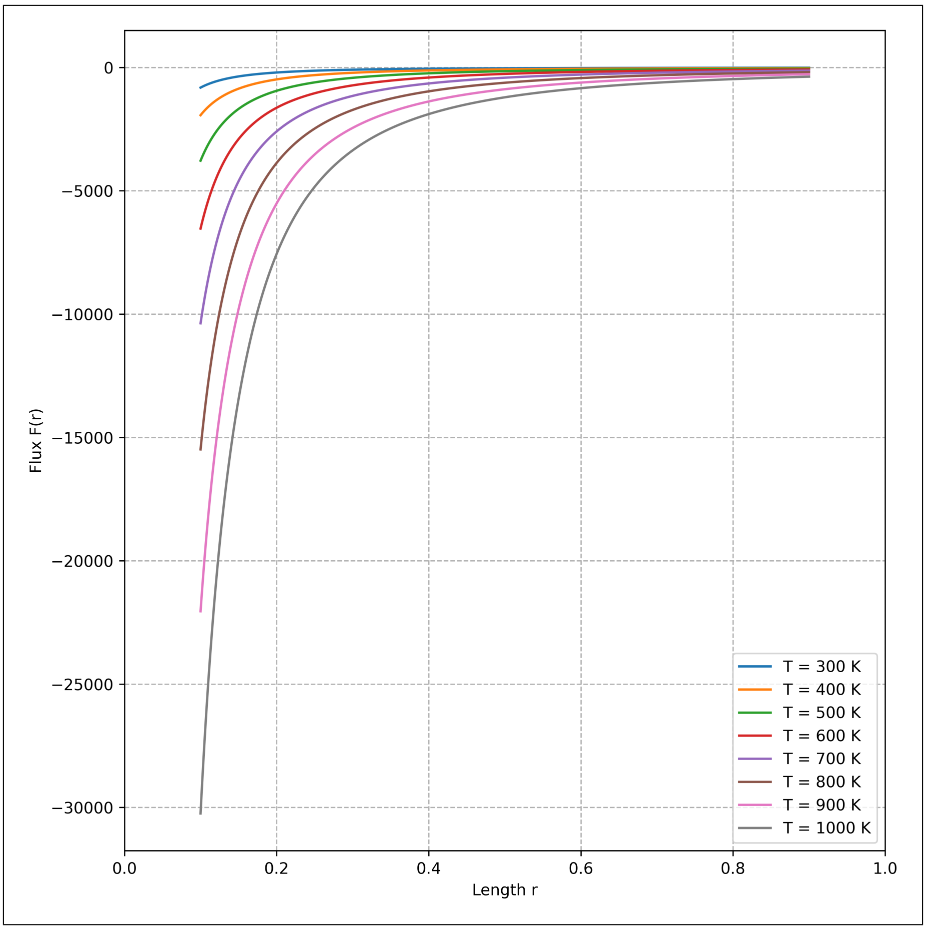

Figure 4: Rosseland approximate energy flux variation with system length parametrised by temperature (Case (a)).

- **Case (b): $\boldsymbol{\alpha_\nu = \alpha_0 r,\ \sigma_\nu = \sigma_0 r}$**

$$F(r) = -\frac{16\sigma_S}{3(\alpha_0+\sigma_0)}\frac{T^3}{r}\frac{\partial T}{\partial r} \tag{35}$$

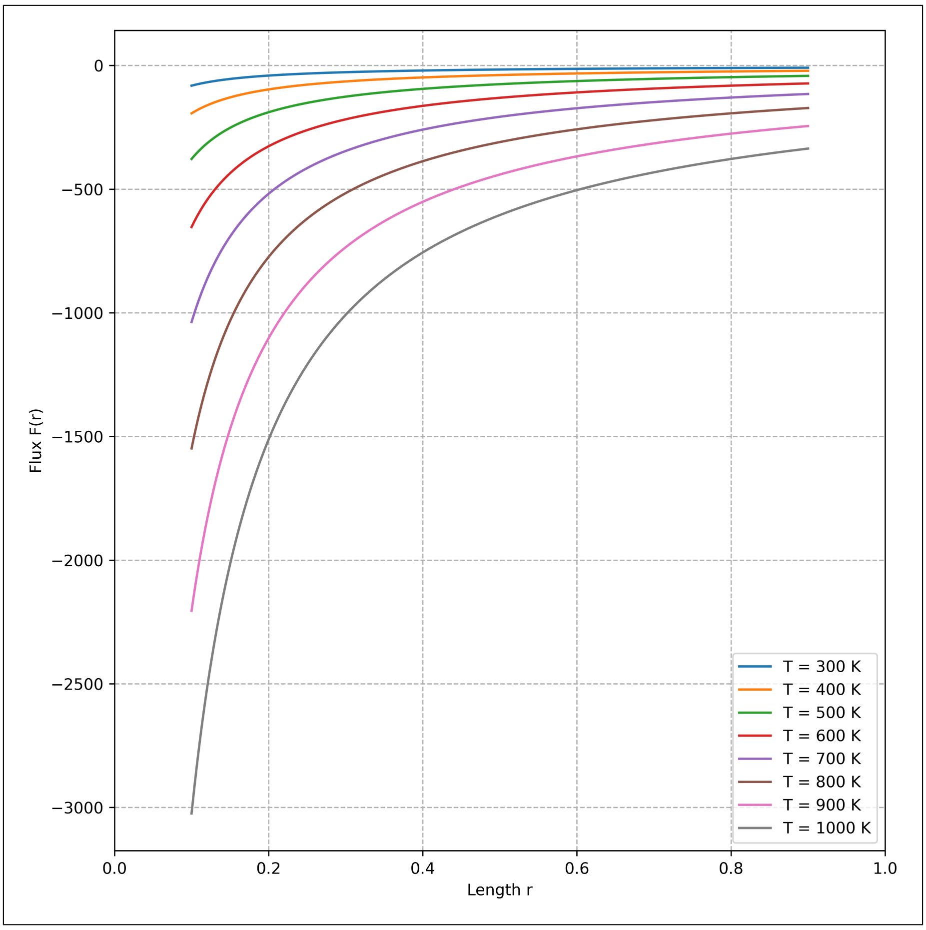

Figure 5: Rosseland approximate energy flux variation with system length parametrised by temperature (Case (b)).

- **Case (c):** $\boldsymbol{\alpha_\nu = \alpha_0 e^{-r/R_0}}$, $\boldsymbol{\sigma_\nu = \sigma_0 e^{-r/R_0}}$

$$F(r) = -\frac{16\sigma_S}{3(\alpha_0 + \sigma_0)} T^3 e^{r/R_0} \frac{\partial T}{\partial r} \tag{36}$$

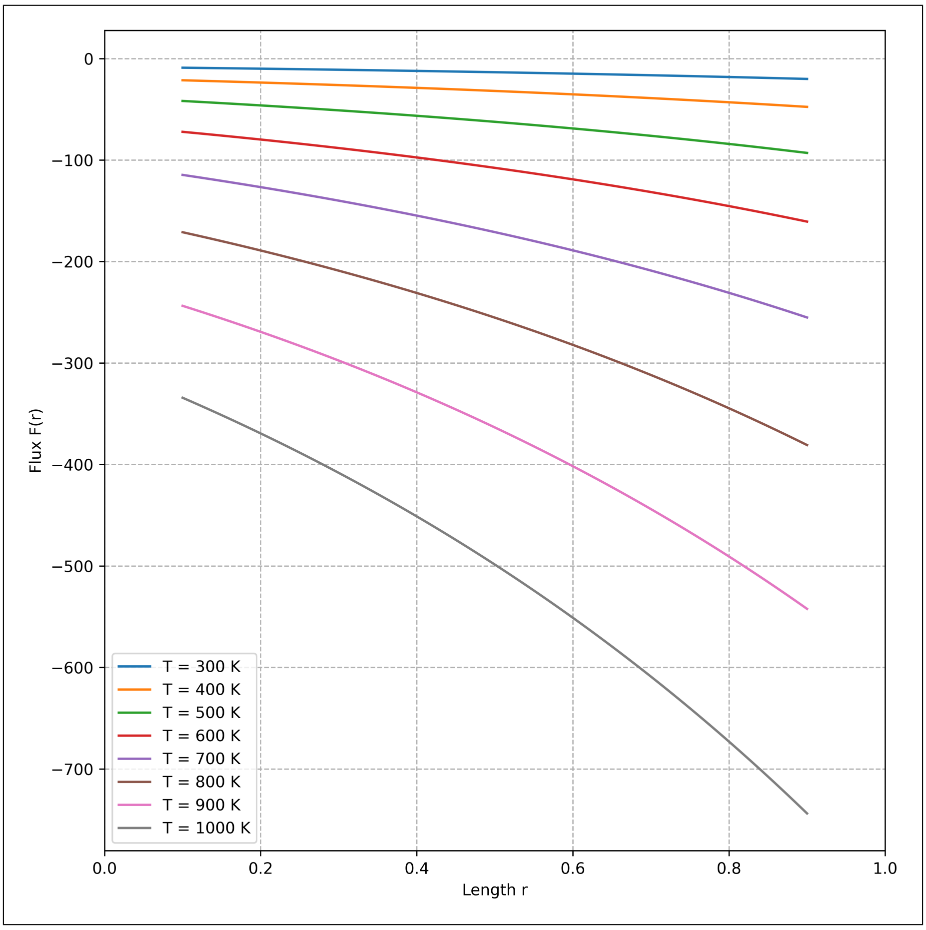

Figure 6: Rosseland approximate energy flux variation with system length parametrised by temperature (Case (c)).

`Links to code:` Question 6(a), Question 6(b), Question 6(c)